Satisfied, Not Solved

What a Demographic Parity Constraint Actually Does

Dheirav

Findings summary — August 13, 2026

Abstract

Fairness constraints of the reduction family (Agarwal et al., 2018) do what they promise: on UCI Adult they drive the demographic parity violation from 0.186 to 0.018 for under two accuracy points. This work audits what they do to *get* there, across **26 populations and 61 experimental arms** spanning a household survey, its modern replacement, and an administrative record of real mortgage decisions.

The central finding: **levelling down is not what parity constraints do — it is what they do below a selection rate of roughly** 0.3. Above it, the same constraint hands decisions out instead of taking them away. This is established by a single-factor design in which only the label's cutoff moves, and replicates across two protected attributes, five states, a second domain, and fourteen populations held out from a later prediction ($r = +0.901$).

1 The problem in one paragraph

A constraint that equalises two groups' selection rates can be satisfied two ways: by extending favourable decisions to the disadvantaged group, or by withdrawing them from the advantaged one. Every real mitigation does a mixture of both. **The certifying metric is identical either way.** It reports that the violation fell; it says nothing about which mixture produced that, or whether the total pool of favourable decisions grew or shrank.

2 What is already known, and what is not

Known. Mittelstadt, Wachter and Russell (2023) established that fairness interventions frequently equalise by degrading the better-off group, and argue it is a default rather than an accident. Their §6 also proposes the remedy — "minimum rate constraints", requiring that every group has at least a minimal selection rate — demonstrated on Adult, and reporting that "levelling down does not occur".

Therefore not claimed here. Neither the observation nor the remedy is novel, and this document says so before a reviewer does. An earlier version of this project claimed the remedy as new; that was wrong about the literature and has been corrected in place.

Not known. *When* it happens. Mittelstadt et al. treat levelling down as a default; their paper contains no discussion of base rates. Their only remark touching prevalence is that Adult is "more than 75% negatively labelled", offered as a note on accuracy cost and never connected to direction. They were operating at a selection rate of roughly 0.24 — inside the levelling-down regime identified here — and did not ask whether it would reverse elsewhere.

The nearest miss. Goethals, Delaney, Mittelstadt and Russell (2024), *Resource-constrained Fairness*, study the *cost* of fairness as a function of the available budget and note the selection rate is "a factor overlooked in previous evaluations". Two of the four authors are also authors of the levelling-down paper. But in their setting positive decisions are a *fixed* resource, so the total cannot move and the directional effect reported here is unobservable by construction. The group best placed to find this result studied the same axis and did not connect it to direction.

3 The central result

3.1 The design

ACS Income's label is "earns more than \$50,000" — a cutoff chosen by the benchmark, not a fact about the world. Moving it on a *fixed* population varies the overall selection rate while holding the population, the survey instrument, the feature set, the group ratio and the proxy structure exactly fixed. Test code asserts that rows, features and protected groups are identical across arms and only the label changes.

The cutoff is the instrument, not the cause. The moderator is the selection rate itself. This is confirmed in a domain where no cutoff exists: HMDA mortgage decisions, where the label is a real lender's approve/deny, sits naturally at a selection rate of 0.808 and lands exactly where the sweep predicts.

3.2 Alabama, six cutoffs, five seeds

Cutoff	Selection rate	Change in favourable decisions	Destroyed per created
\$100,000	0.030	−29.71%	22.03
\$70,000	0.099	−22.05%	2.18
\$50,000	0.252	−2.34%	1.14
\$30,000	0.598	+0.98%	0.83
\$20,000	0.760	+0.80%	0.75
\$10,000	0.890	+0.08%	0.89

Same people. Same features. Same groups. Only the finish line moved — and the direction flipped, with the crossover between 0.25 and 0.60.

3.3 Replication

Evidence	Scope	Result
Original sweep	Alabama, Oregon (sex)	$r = +0.801$, $+0.964$
Second protected attribute	4 states (race)	$r = +0.874$ to $+0.991$
Held-out populations	14, run after prediction	$r = +0.901$
Second domain	HMDA, 2 states, 2 arms	correct side of crossover
Datasets never fitted to	Adult (0.205), HMDA (0.808)	both placed correctly

3.4 The obvious objection, pre-empted

That unequal base rates *between groups* force trade-offs is textbook (Kleinberg et al., 2016; Chouldechova, 2017). **This claim is about a different quantity**: the overall *level* of the selection rate, irrespective of any gap.

The two are confounded by construction — the between-group gap must vanish as the level approaches 0 or 1. So the gap is partialled out, and the relationship *strengthens*: from $r = +0.801$ to $+0.980$ in Alabama, and $+0.964$ to $+0.994$ in Oregon. That control is the identification strategy, not a robustness check.

4 An identity that says precisely what levelling down is

Let the privileged group be share p with unconstrained selection rate r_A, and the unprivileged group $1-p$ at r_B. The unconstrained total is the size-weighted average $\bar{r} = p\,r_A + (1-p)\,r_B$. Demographic parity puts *both* groups on one common rate s, so the constrained total *is* s, and

$$\text{change in favourable decisions} \;=\; s - \bar{r}.$$

Writing $\lambda = (s - r_B)/(r_A - r_B)$ for where the common rate falls between the two original rates,

$$\boxed{\text{the pool is preserved exactly when } \lambda = p.}$$

Levelling down is $\lambda < p$ and nothing else: the optimiser choosing a compromise *below* the size-weighted one. This requires no assumptions, holds to 0.076 percentage points across fourteen held-out populations, and does not appear to be stated anywhere in the literature reviewed.

5 What could not be established

The mechanism. A derivation was attempted: under group-wise thresholding, calibration and a location shift, levelling down is a *curvature* effect and the crossover should sit at the mode of the score density. It was pre-registered with numerical thresholds and tested on fourteen populations run afterwards. It cleared its bars (predicting the sign in 12 of 14) and was **beaten by the constant rule "levelling up iff the selection rate exceeds 0.5" (13 of 14)**. The measured mode moves inversely with the selection rate ($r = -0.802$), so it carries almost no independent information.

The methodological lesson is recorded with it: a *threshold* was fixed in advance, but a *baseline to beat* was not, so clearing the threshold did not mean what it appeared to mean.

The magnitude. The selection rate sets the direction, not the size. Alabama loses 2.34% at a rate of 0.252 where Adult loses 20.5% at 0.205. A pre-registered candidate — the group ratio — was refuted, with the opposite sign to the prediction.

6 Supporting findings

Subgroups. A constraint on one attribute leaves Sex × Race subgroups worse off than any group it was told about, in every sufficiently diverse population, and more severely than Adult suggests (9.0× against 13.2×). In five of ten populations the worst-off subgroup after a sex constraint is a minority man.

Attribution audits do not reveal the mechanism. A parity constraint moves SHAP attribution *onto* a sex proxy by +151%. Three explanations were proposed; two were refuted by intervention and the third bounded by re-aggregation — scoring the two most redundant features as one coalition takes the effect from +155% to +11.6%. The transferable lesson: **an attribution audit can move by an order of magnitude without the model's behaviour changing correspondingly.**

Arbitrariness. Below roughly 2,500 test subjects, the method's own randomness exceeds the entire effect of the constraint.

Against existing remedies. Group-wise thresholding — the *optimal* DP-constrained classifier — destroys 18.8% of favourable decisions on Adult against the reduction's 20.5%, so levelling down is not a solver artifact. Minimax group fairness, proposed in the literature as *the* remedy for levelling down, destroys 10.0% of favourable decisions on HMDA at an exchange rate of 46.7 — on the same data where the parity constraint *creates* 4.3%. Its objective minimises the worst group's *error*, and worst-off-by-error is not worst-off-by-outcome: on Adult the higher-error group is Male.

7 Why it matters

Lending, hiring and admissions — the domains where these methods are actually proposed — are nearly all low-selection-rate settings. **The default regime for deployed fairness constraints is therefore the one in which they take favourable decisions away, and the fairness score never says so.**

8 Method, and what it caught

Five seeds throughout; splits stratified on the (attribute, label) interaction; the protected attribute removed from the feature matrix; metrics implemented from their definitions and cross-checked against `fairlearn` on every run; undefined rates returned as undefined rather than zero.

Where a result could have gone either way, predictions and their numerical thresholds were written and committed *before* the experiment ran, verifiable in the commit history. That discipline is why the following can be reported: **three pre-registered predictions of my own failed**; two proposed mechanisms were refuted by my own interventions; and three claims were corrected or retracted in place, including the novelty claim for the remedy, which the literature had anticipated.

9 The open question

Is the scope condition sufficient on its own for a venue such as FAccT or AIES, or should the mechanism be established first? The derivation attempted here failed, and is reported as a failure. The finding is *when* the direction flips, not *why*.

All figures are reproduced from stored results by `tests/test_documented_claims.py`, which re-derives every headline number and fails if a document and its data disagree. Supporting papers are in `papers/`; the full research record is in `research/docs/`.